Type MS · Explain red-cell adaptations and the effects of more or less oxygen carriage

Answer 1 of 5 · Question 1 part a.ii · 4 marks

2021 NOV · Paper 02 · extract 1 of 2 · find part a.ii in this mark scheme extract

Question number	Answer	Notes	Marks
1 (a) (i)	• transport oxygen; • from lungs; • to (body) tissues/cells;	Allow organs	3
(ii)	• haemoglobin to carry/combine with oxygen; • no nucleus, so more haemoglobin; • biconcave shape to increase surface area; • small/flexible (cell) membrane to pass through capillaries; • thin membrane to reduce diffusion distance;		4
(b) (i)	vein;		1
(ii)	• valve present/large lumen/thin (muscular) walls; • only found in veins/not found in arteries/other blood vessels;		2
(iii)	• distance 13-15 mm; • divide by 10 = 1.3-1.5;	Ecf: max 1 mark if mp1 incorrect but division by 10 of their value is correct	2
			Total 12

Question number	Answer	Notes	Marks

Type MS · Explain red-cell adaptations and the effects of more or less oxygen carriage

Answer 1 of 5 · Question 1 part a.ii · 4 marks

2021 NOV · Paper 02 · extract 2 of 2 · find part a.ii in this mark scheme extract

Type MS · Explain red-cell adaptations and the effects of more or less oxygen carriage

Answer 2 of 5 · Question 5 part b.iii · 5 marks

2023 JAN · Paper 02 · extract 1 of 2 · find part b.iii in this mark scheme extract

Question number					
5	(a)	(i)	$C_6H_{12}O_6$; $6H_2O$;		2
		(ii)	ATP breaks down/hydrolysed; to ADP and Pi;	Allow phosphate/P	2
	(b)	(i)	• suitable axis labels with units; • time on x axis; • suitable scale; • correct plots;		4
		(ii)	• suitable curve;		1
		(iii)	Five from; • mass of haemoglobin increases (with time/training at altitude); • increased number of red blood cells/more haemoglobin in the blood; • haemoglobin binds oxygen/red blood cells carry oxygen; • more oxygen transported around the body/in the blood; • to muscle (cells/tissue); • for aerobic respiration; • to release energy;		5

Total 14 marks

Type MS · Explain red-cell adaptations and the effects of more or less oxygen carriage

Answer 2 of 5 · Question 5 part b.iii · 5 marks

2023 JAN · Paper 02 · extract 2 of 2 · find part b.iii in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Explain red-cell adaptations and the effects of more or less oxygen carriage

Answer 3 of 5 · Question 6 part a.ii · 5 marks

2023 JUN · Paper 02 · extract 1 of 2 · find part a.ii in this mark scheme extract

Question number				
6	(a) (i)	• change in genetic code/base sequence/nucleotide sequence/different codon; • of DNA/gene; • different amino acid coded for/different sequence of amino acids; • different (shaped) protein produced;	Allow named mutation e.g. deletion	4
	(ii)	Any five from • shape of red blood cell distorted; • different form of haemoglobin; • reduced surface area; • less oxygen carried; • less respiration; • less energy/ATP;	Allow red blood cells are sickle-shaped Allow capillaries blocked/allow less haemoglobin	5
	(b)	• woman is Hh; • man is either Hh or hh/heterozygous or homozygous recessive; • one h/recessive allele comes from man; • child is hh/homozygous recessive;		4

Total 13

Type MS · Explain red-cell adaptations and the effects of more or less oxygen carriage

Answer 3 of 5 · Question 6 part a.ii · 5 marks

2023 JUN · Paper 02 · extract 2 of 2 · find part a.ii in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Explain red-cell adaptations and the effects of more or less oxygen carriage

Answer 4 of 5 · Question 5 part b.i · 4 marks

2024 NOV · Paper 01 · extract 1 of 2 · find part b.i in this mark scheme extract

5 (a)	(i)	• 13 750 ÷ 7500; • 1:1.83;	Allow 1:1.8/1:2/ 1:1.8333... allow if they do it the other way round ie 7500 ÷ 13750	2
	(ii)	• person A has an infection/disease; • Immune response/more white blood cells needed, to produce antibodies/phagocytosis/destroy pathogen/virus/bacteria;	ignore sick ignore fight infection allow alternative correct explanation	2
	(iii)	Any two from: • excess bleeding; • poor/no blood clotting; • poor/slow wound healing/more time for pathogens to enter wound;		2
(b)	(i)	• tiredness/fatigue/high heart rate/faster breathing; • less oxygen carried/transported (to body cells); • less aerobic respiration; • less energy;		4
	(ii)	• HA is dominant; • parents are heterozygous; • HA expressed in heterozygous form; • so only needs one allele;	ignore ref to allele/gene for mpt 1	4

Total Marks 14

Type MS · Explain red-cell adaptations and the effects of more or less oxygen carriage

Answer 4 of 5 · Question 5 part b.i · 4 marks

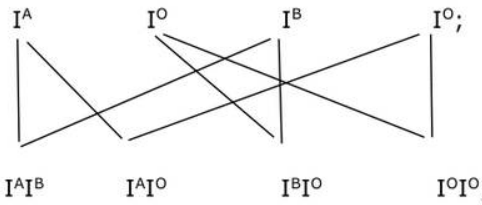
2024 NOV · Paper 01 · extract 2 of 2 · find part b.i in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Explain red-cell adaptations and the effects of more or less oxygen carriage

Answer 5 of 5 · Question 4 part b · 4 marks

2022 JUN · Paper 01 · extract 1 of 7 · find part b in this mark scheme extract

4 (a)	(i)	**Africa** Blood group / Percentage of blood group O: 68 A: 17/17.5 B: 10; AB: 5 **Europe** Blood group / Percentage of blood group O: 45 A: 40 B: 12 AB: 3;		2
	(ii)	• reference to antigens on red blood cells (group B); • antibodies produced; • causing agglutination/clumping (of red blood cells);		3
	(iii)	I^AI^O I^BI^O;  I^A I^O I^B I^O; I^AI^B I^AI^O I^BI^O I^OI^O;	Full marks for a Punnett square if parent genotypes are given.	3
	(iv)	1 in 4:/25%/¼/0.25;		1
(b)		• <u>deoxygenated</u> blood moves into the left side/ventricle of the heart; • (deoxygenated blood) mixes with <u>oxygenated</u> blood; • less oxygen travelling around the body/to body/cells/in bloodstream; • breathes faster to get more oxygen into the body;		4
				Total 13

Type MS · Explain red-cell adaptations and the effects of more or less oxygen carriage

Answer 5 of 5 · Question 4 part b · 4 marks

2022 JUN · Paper 01 · extract 2 of 7 · find part b in this mark scheme extract

	Answer	Notes	Marks

Type MS · Explain red-cell adaptations and the effects of more or less oxygen carriage

Answer 5 of 5 · Question 4 part b · 4 marks

2022 JUN · Paper 01 · extract 3 of 7 · find part b in this mark scheme extract

Question number	Answer	Notes	Marks
6	Any six from the following: • remove plasmid from bacteria; • locate the human insulin gene • cut/remove insulin gene (from human); • cut open the plasmid; • using restriction enzymes; • add the human insulin gene to the plasmid; • use of ligase; • ref recombinant DNA; • return the plasmid to a bacterial cell;	Suitably drawn **and** annotated diagrams can score full marks.	6
			Total 6

Question number	Answer	Notes	Marks
7 (a) (i)	obtain a pure product/enzyme and product not mixed/enzymes are reusable/active over wider range of temperature/pH;		1
(ii)	sucrose;		1
(iii)	any three from • add sodium alginate with distilled water; • mix (with a hand blender); • until all sodium alginate has dissolved; • add calcium chloride; • wash beads;		3
(b)	**B** insulin is released from the pancreas and travels to the liver **A** is incorrect as glucagon does not convert glucose to glycogen **C** is incorrect as glucagon is not released from the liver **D** is incorrect as insulin is not released from the liver		1]

Type MS · Explain red-cell adaptations and the effects of more or less oxygen carriage

Answer 5 of 5 · Question 4 part b · 4 marks

2022 JUN · Paper 01 · extract 4 of 7 · find part b in this mark scheme extract

(c) (i)	• (66.65 ÷ 100) = 0.6665; • (x6) = 4.0 (million);	Allow 0.67 Allow 3.999	2
(ii)	• 38 49 51 62 65 78 82 91 96; • 65;		2
			Total 10

Type MS · Explain red-cell adaptations and the effects of more or less oxygen carriage

Answer 5 of 5 · Question 4 part b · 4 marks

2022 JUN · Paper 01 · extract 5 of 7 · find part b in this mark scheme extract

Question number	Answer	Notes	Marks
8 (a)	**Advantages embryonic stem cells:** • generate most cell types; • can maintain and grow for long lengths of time in culture; • useful for drug development studies/Parkinson's disease; **Disadvantages of embryonic stem cells:** • can cause tumours/cancer; • many embryos destroyed; • some people think it is unethical/killing/murder; • possibility of rejection if used in transplants **Advantages of adult stem cells:** • can be reprogrammed; • less likely rejected if used in transplants; • taken from the patient itself; **Disadvantages of adult stem cells:** • limited ability to differentiate/develop into different types of cell; • cannot be grown for long periods of time in culture; • usually a very small number found in body tissues; • can only grow small numbers;	Three marks from embryonic stem cell section and three marks from adult stem cell section Allow a valid named ethic	6
			Total 6

Type MS · Explain red-cell adaptations and the effects of more or less oxygen carriage

Answer 5 of 5 · Question 4 part b · 4 marks

2022 JUN · Paper 01 · extract 6 of 7 · find part b in this mark scheme extract

Question number	Answer	Notes	Marks
9 (a)	• beta blockers; • block/bind to receptors; • which prevents the release of adrenaline/noradrenaline; • heart rate decreases; OR • ACE inhibitors; • reduce the activity of enzymes; • less angiotensin; • less narrowing of blood vessels;		4
(b)(i)	any two from • how much pressure blood is exerting; • against artery walls; • when ventricles contract/pump;		2
(ii)	• pressure in the arteries; • when the heart/ventricles at rest/relaxed/between beats;		2
			Total 8

Type MS · Explain red-cell adaptations and the effects of more or less oxygen carriage

Answer 5 of 5 · Question 4 part b · 4 marks

2022 JUN · Paper 01 · extract 7 of 7 · find part b in this mark scheme extract

Pearson Education Limited. Registered company number 872828
with its registered office at 80 Strand, London, WC2R 0RL, United Kingdom